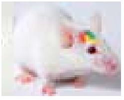
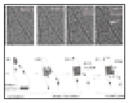


Wireless brain implant
Using ultra-soft and bio-compliant polymers, scientists created a smartphone-controlled soft brain implant. **http://dgit.in/feb21-23**



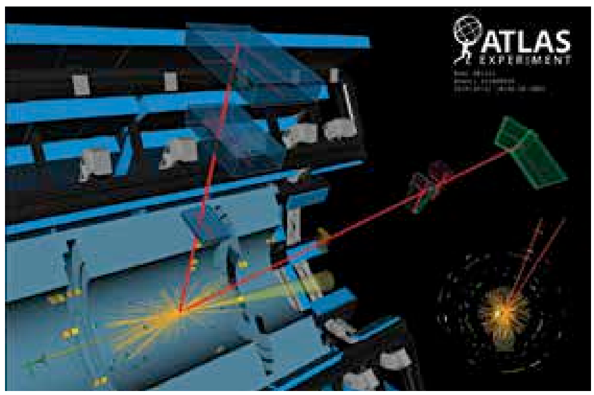
Crystal close up
First-ever atomic resolution video reveals formation of salt crystals from individual atoms in real time. **http://dgit.in/feb21-24**

model. The interactions with heavier particles were better understood, but the interactions with the lighter particles were rarely seen, and these were the ones that interested scientists the most. While we do know that the Higgs boson provides all particles in the universe with their mass, allowing gravity to exist, we do not fully understand why some particles are more massive than others. It was predicted that the Higgs field would also be able to provide mass to extremely light fermions (quarks and leptons), through interactions called "Yukawa couplings". In 2018, Atlas detected the Higgs boson decaying into a pair of tau leptons, providing the first direct evidence of Yukawa couplings. 60 percent of all Higgs boson decays were expected to be into a pair of beauty (also known as bottom) quarks, which was first detected in 2018 as well. The discovery moved the whole area of research around Higgs boson from the discovery era to the measurement era. This is because it was now necessary for scientists to observe the decay mode into two b quarks, which is the ideal mode to observe because of the large probability it would take place, and measure its precise rate to confirm or exclude Yukinawa interactions as the process by which fermion mass generation occurs. Measuring the deviations in different production mechanisms of the Higgs boson, as well as their decays allows scientists to fine tune their understanding of physics, or discover entirely new phenomena beyond the standard model.

You might be wondering if 60 percent of all the Higgs boson decays produce a pair of b quarks, why are they so hard to discover? This is because the best way we know to produce the Higgs boson, proton to proton interactions, results in a pair of b quarks most of the time. It is almost overwhelmingly difficult to isolate the signature of the Higgs boson from this background noise. As you can imagine, the researchers need large volumes of detection events to come up with the data that has statistical significance. The good thing about this approach is that having a detailed record going back years allows researchers to apply analysis techniques on existing data, and find events that had previously eluded them. A search was undertaken for odd events that would result in a pair of b quark decays, such as when the Higgs boson is produced at an angle that is extremely perpendicular to the proton beams. In such cases, the two b quarks form jets next to each other, instead of on opposite ends of the detector ring. An event in 2017 discovered in a later analysis is shown in the title image.

SUPERCOMPUTERS AND AI

There are a few main decay channels or modes of the Higgs boson, into pairs of photons, W bosons, Z bosons, tau leptons, b quarks and theoretically, muons as well as dark matter. The scientists observe and measure all of these events to understand new physics. The two Z bosons can further decay into four neutrinos, that pass undetected through the LHC experiments. It is also theoretically possible for the Higgs boson to decay into dark matter particles, especially if the elusive dark matter particles have a mass that is less than half of the Higgs boson. All known massive particles interact strongly with the Higgs boson, and scientists see no reason why the same should not hold true for dark matter particles as well. However, differentiating between events where Higgs boson decays into dark matter, and the invisible decays into neutrinos could be extremely difficult, which is where the scientists are using statistical methods again, with progress being made on fine tuning the constraints needed to detect interactions involving dark matter in the future. In 2020, CERN scientists released new findings that used adversarial neural networks, to isolate the b quark events from the overwhelming background noise. While there were many potential decays into a pair of muons, this last event had remained elusive and unconfirmed till October 2020, when researchers from Berkeley lab found evidence that strongly supports the hypothesis of the Higgs boson decaying into a pair of muons. These are the lightest particles yet known to interact with the Higgs boson. The scientist had to extract a very tiny signal hiding in all the noise. To help with the background estimation of detection events, the Berkeley lab scientists used several supercomputers to simulate 10 billion particle interaction events. These simulations allowed the researchers to understand the theoretical values for the events actually observed at CERN. By comparing the actual experiments to the simulations, physicists can figure out the odd balls, identify deviations, and isolate a signal. After the simulations, machine learning techniques were used to isolate each type of Higgs boson event from the background noise. In all, the system managed a 25 percent improvement in sensitivity to discovering muon decays. Based on the study, a planned upgrade to the LHC should be able confirm the muon decays.



Credit: ATLAS/CERN

A candidate event of the Higgs boson decaying into a pair of muons